TRAINING-FREE · NO TASK HEAD · OPEN VOCABULARY

Test-time compute of jina-embeddings-v5-omni for image tagging

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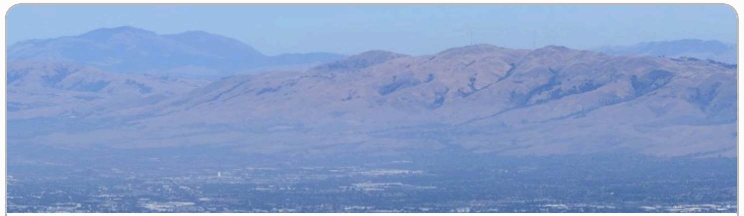
GITHUB REPO

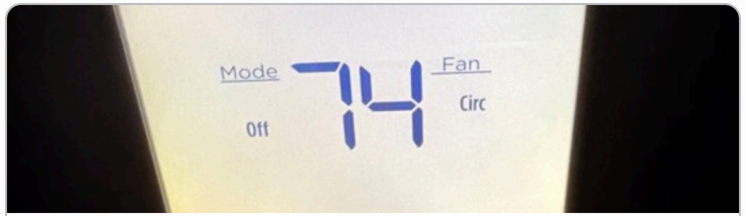
code · benchmark · findings

THE RESULT, FIRST

Training-free, open-vocabulary tagging from a frozen retrieval model.

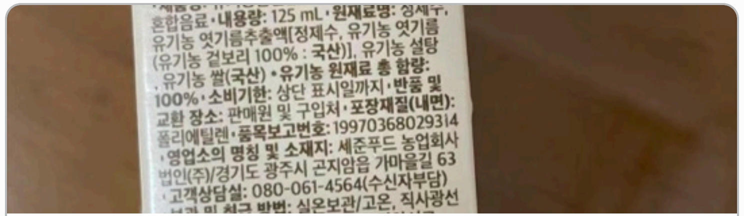


mountain · phoenix · sky · rooftop · location

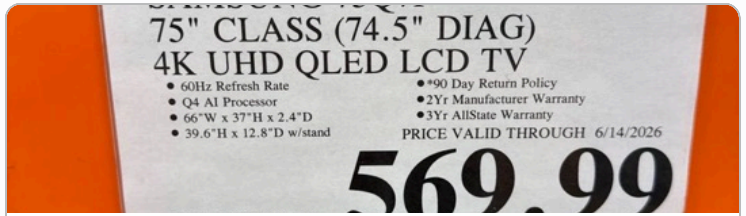


clock · thermostat · dial · dst · wifi

fiyat · pagamento · brasile · gratuiti · utils

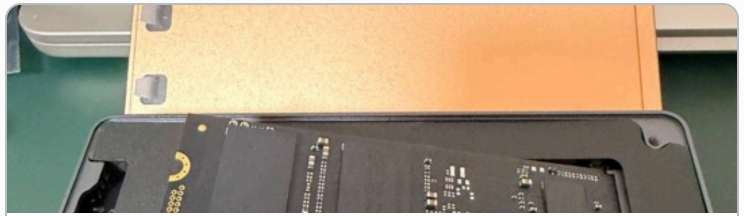


contiene · liters · dosage · thirst · consumo

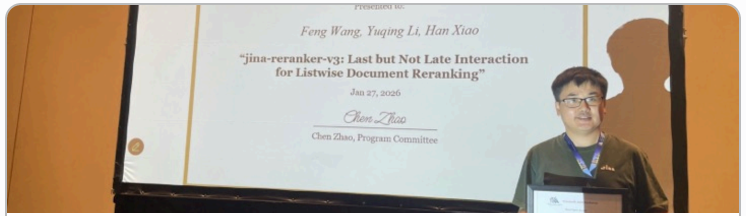


harga · lcd · samsung · refurbished · warrants

sunglasses · watches · branded · bronze · necklace



tablet · motherboard · wearable · celular · datastore



award · forwarding · speaking · defendant · workshop

gift · toy · celebration · balloons · cheerful

Unstaged files from this machine; top-5 tags, unfiltered; 75 ms each. P@1 0.813 on COCO-150. The rest of the talk: where this comes from, and which compute pays for it.

SETUP

Problem formulation.

GIVEN one embedding model, **jina-embeddings-v5-omni-nano** (~1B, image and text in one 768-d space), and one image.

OUTPUT words for **every** object in the image: multi-label, open vocabulary.

FROZEN all weights. The model is a retrieval encoder; it has no tagging head, no classifier, and was never trained for this task.

ALLOWED inference-time computation over the model's own outputs, and nothing else.

TRAINING-FREE | NO SECOND MODEL | NO WORDNET / DICTIONARY | NO REGEX / POS TAGGER

Whatever tagging ability appears must come from test-time compute of the embedding model.

A TAXONOMY

Where new information comes from.

A frozen encoder's inference-time budget takes roughly three forms. Superficially all three are "more math on the model's outputs"; they differ on one axis: whether new information enters the computation, and from where.

A READ MORE OF THE PASS

The pass computes a vector per token or patch; single-vector use throws them away. Reading them is the retrieval world's **late interaction** (ColBERT-style rescoring).

new information: reclaimed from inside the pass

B RUN NEW PASSES

Encode inputs the model has not seen: split a document and embed the pieces, crop an image and re-encode. New forward passes over new views.

new information: acquired from the input

C RE-ARRANGE WHAT YOU HAVE

Transform a **fixed** set of vectors: whitening, graph propagation, optimal transport, re-normalization. Most of the recent training-free literature lives here.

new information: none

Same frozen encoder in all three; they differ only in what enters the computation.

RESEARCH QUESTIONS

Two research questions.

RQ1 **Can a frozen embedding model become an image tagger through test-time compute alone?**

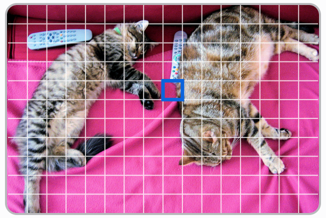
Prior tagging work pays with training or resources: Tag2Text / RAM / RAM++ train over ~6,400 curated tags; TagCLIP (AAAI'24) and PIAA ("[CLS] is not enough") are training-free but assume a given class list, on two-tower CLIP plus WordNet or POS tools.

RQ2 **If it can: which test-time compute scales its accuracy, re-processing features or new information?**

The 2024–26 literature bets on family C: whitening (PIAA), label propagation (ZLaP), optimal transport (OTTER), Bayesian adaptation (BCA). Each is measured here against families A and B.

THE MODEL · ARCHITECTURE

One frozen checkpoint, two input paths, one 768-d space.



PER IMAGE · 16 PX PATCHES · TEST TIME

cat

coche

橛

× 128,260 tokens

VOCABULARY · ENCODED ONCE

jina-embeddings-v5-omni-nano · ~1B · frozen

Qwen3-VL vision tower
16 px patches
2×2 merge

EuroBERT-12L text tower
bidirectional · 768-d

vision tokens injected at image-token positions

never modified, never retrained

P: one 768-d row per patch (the tower's internal 16 px unit), each vector shaped by attention over the *whole* image

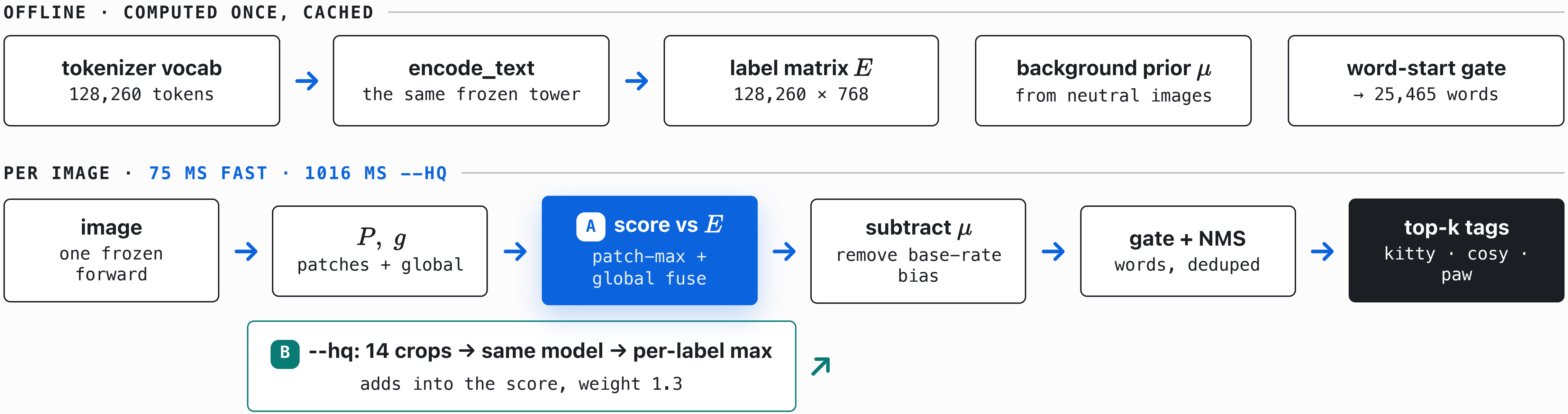
g: one vector for the whole image, pooled from the last token of the same sequence

E: label matrix, 128,260 × 768; gated to 25,465 words at scoring

Both paths end in the same space, so $\langle p,\ e_\ell \rangle$ is well-defined: any patch of any image, scored against any word the model knows.

THE PIPELINE

The complete tagger: test-time compute around a frozen forward pass.



Every box is the frozen model or plain arithmetic: A reads more of one pass; B runs new passes on new pixels.

STEP 1 · THE LABEL SPACE

The label set is the tokenizer's own vocabulary.

No curated tag list. Take all **128,260** tokens and encode each one as text into a label matrix

$$E = \left[\, e_\ell \,\right]_{\ell \in V}, \qquad e_\ell = \mathrm{encode\_text}(\ell) \in \mathbb{R}^{768}, \quad |V| = 128{,}260$$

One matmul now scores any image against the entire vocabulary; fragments are filtered later by a word-start gate (step 4).

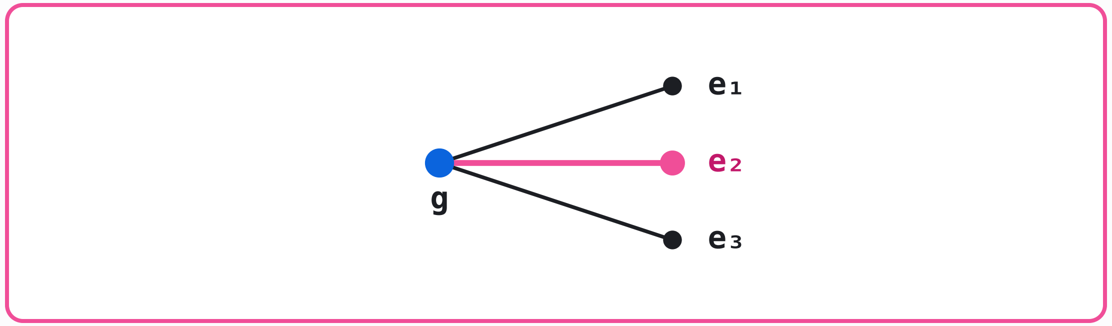
```
wrong: image_vec · embed_tokens.weight
      → Tutor, avatar, PyTuple  (garbage)

right: image_vec · encode_text(token)
     → kitty, cosy, plush, paw  (aligned)
```

Why encode_text, not the embedding table? The model has `tie_word_embeddings = false`: the input-embedding table lives in a different space than the pooled output. You must encode each token *as a text string* to put labels in the image's space.

STEP 2 · WHERE TO LOOK

Score every patch.

GLOBAL POOLED

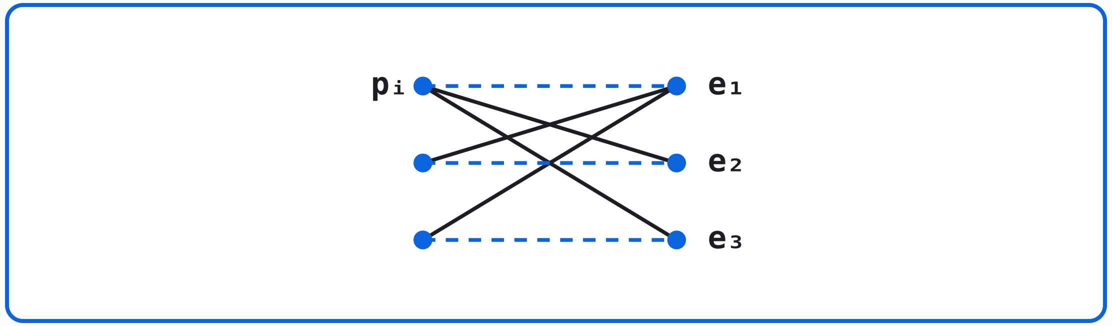


$$s_\ell = \langle g, e_\ell \rangle$$

one vector dominated by the most salient object

mAP 0.264

PER-LABEL MAX OVER PATCHES



$$s_\ell = 0.7\,\max_{p\in P}\langle p, e_\ell \rangle + 0.3\,\langle g, e_\ell \rangle$$

every label takes its max over all patches: late interaction (ColBERT), with patches as the token vectors

mAP 0.635

A PIAA's "[CLS] is not enough", confirmed: a small object fires on its own patch and survives the max. +0.37 mAP, from rows the pass already produced.

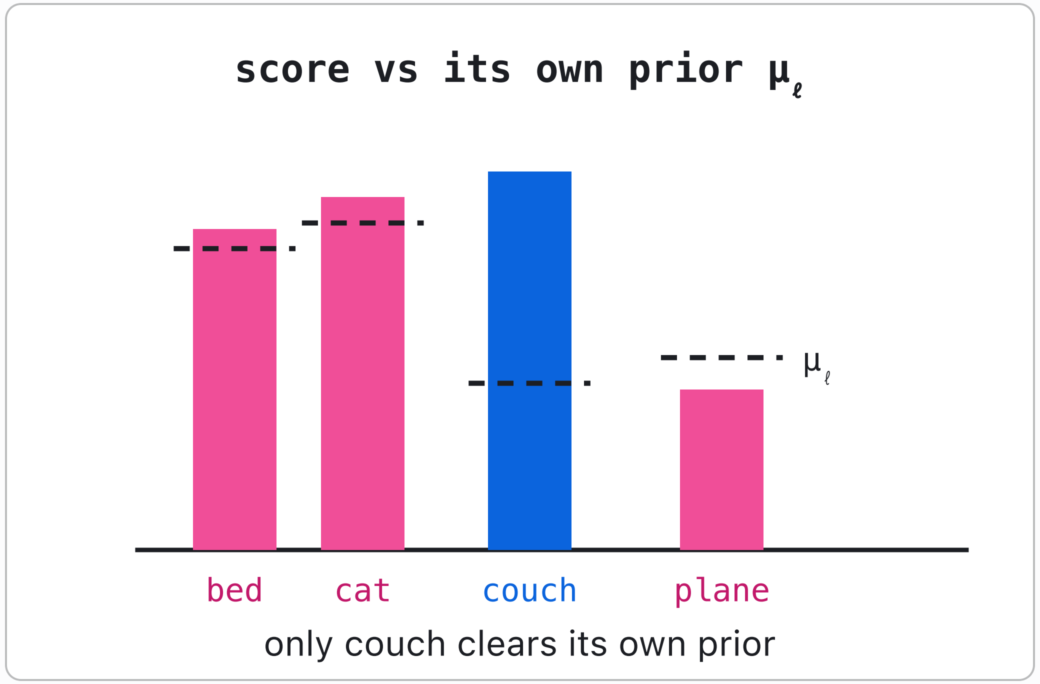
STEP 3 · DE-BIAS

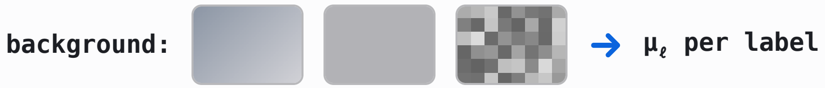
Subtract the per-label prior.

Raw cosine has a base-rate bias: generic words ("bed", "cat") score high on almost any image because they sit near the image modality. Estimate a per-label prior once from neutral background images, then subtract it:

$$\tilde{s}_\ell \;=\; s_\ell - \mu_\ell, \qquad \mu_\ell = \mathbb{E}_{x\,\sim\,\text{background}}\big[s_\ell(x)\big]$$

No calibration set, no labels, one offline pass.





STEP 4 · CLEAN UP

The tokenizer already knows what a word is.

Word-start gate. Byte-BPE tokenizers encode a leading space as the glyph `Ġ`; a token beginning with it starts a new word: `Ġcat` is a whole word, `GetComponent` is a fragment. That one built-in signal is the gate; no external dictionary.

Embedding-NMS. Synonyms and multilingual variants (/ Cat / кот / cat) collapse into one, using cosine in the model's own space. Walk labels by score; keep ℓ only if $\max_{k \in \text{kept}} \langle e_\ell, e_k \rangle < 0.6$. Again: no lookup table, just the geometry.

raw vocabulary — 128,260 tokens

`Ġkitty` · `Ġcat` · ~~`GetComponent`~~ · ~~`_cpp`~~ · 猫 · ~~`quisites`~~

↓ word-start gate: keep tokens with the leading-space mark Ġ

whole words — 25,465

`kitty` · `cat` · `kitten` · `chatte` · `kitt` · `cats` · `kittens` · `cosy` · …

↓ embedding-NMS: walk down by score; drop any word with cosine ≥ 0.6 to one already kept

distinct concepts — top-k

`kitty` · ~~`cat`~~ · ~~`kitten`~~ · ~~`chatte`~~ · ~~`cats`~~ · `cosy` · `plush` · `crib`

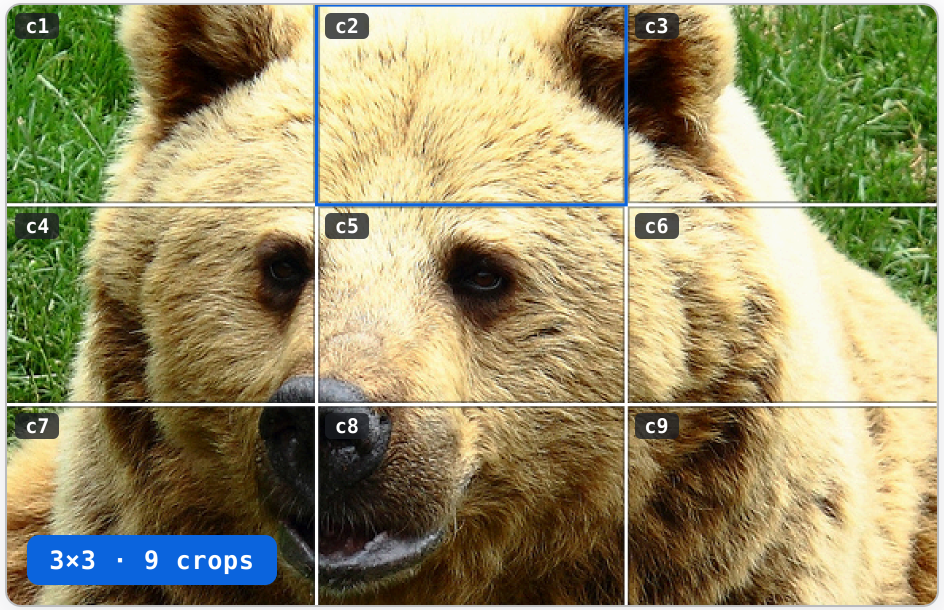
STEP 5 · NEW VIEWS

Multi-crop re-encoding.

Class-Wise Reranking (CWR), the crop-and-rescore idea from TagCLIP (AAAI'24), extended to a full-coverage grid of 14 crops (overlaid right) through the same frozen model. A crop is a *new input image*, itself re-patchified by the tower; a small object fills whichever crop contains it, so weak signal becomes strong.

$$S_\ell = \tilde{s}_\ell + 1.3\Big(\max_{c=1..14} s_{c,\ell} - \mu_\ell\Big), \qquad s_{c,\ell} = \max\Big(\max_{p\in P_c}\langle p, e_\ell\rangle,\ \langle g_c, e_\ell\rangle\Big)$$

B Per-label **max**, not averaging: averaging hurt. The outlier crop is the evidence.



fast: wolf, roar, muzzle → --hq: fur, **bear**, muzzle
multi-crop re-encoding corrects the single-pass misreading

ALL FIVE STEPS, ASSEMBLED

The entire tagger is one scoring function.

$$S(\ell) \;=\; \underbrace{0.7\left(\max_{p\in P}\langle p,\, e_\ell\rangle - \mu_\ell\right)}_{\textbf{patch evidence}} + \underbrace{0.3\left(\langle g,\, e_\ell\rangle - \mu^g_\ell\right)}_{\textbf{global context}} + \underbrace{1.3\left(\max_{c=1..14} s_{c,\ell} - \mu_\ell\right)}_{\textbf{multi-crop re-encode}}$$

A **Patch evidence**: algebra over rows the forward pass already produced. Free. +0.37 mAP.

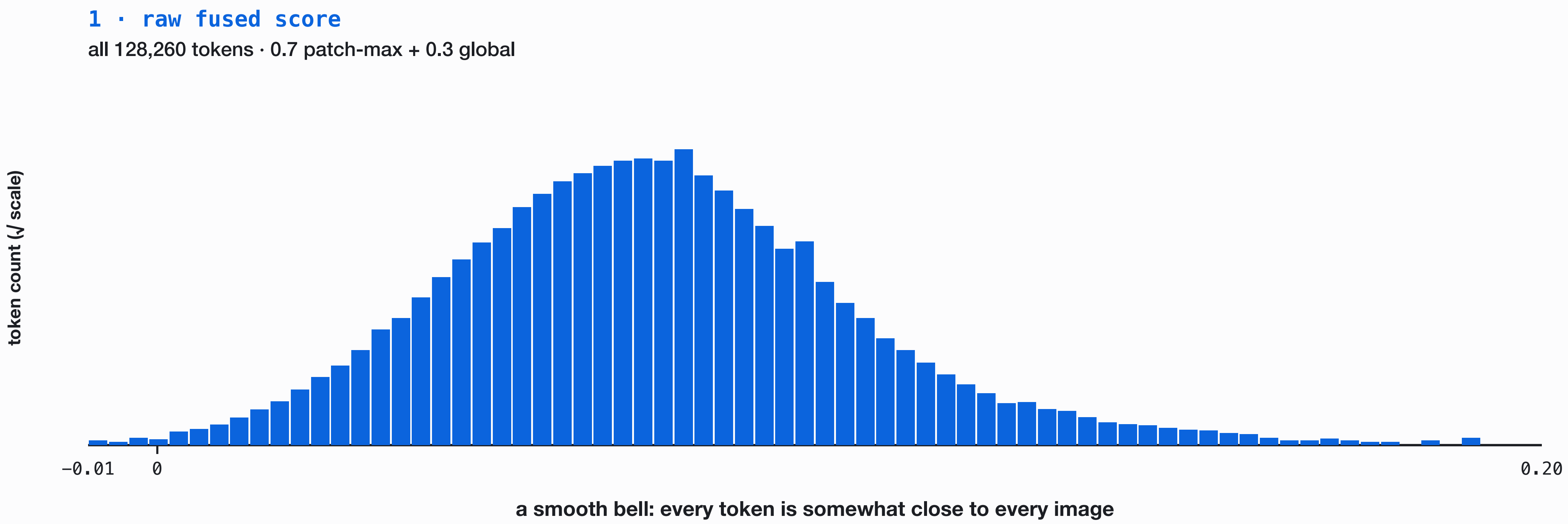
A **Global context**: the pooled vector, de-biased by its own background prior. Free.

B **Multi-crop**: 14 fresh forward passes on new pixels. 14× the cost. +0.075 mAP.

where $s_{c,\ell} = s_\ell(\mathrm{crop}_c)$: **the same score, recursively applied to each crop** · then $\ell \in \text{word-gated vocabulary} \rightarrow \text{embedding-NMS } (\tau = 0.6) \rightarrow \text{top-}k$

No head, no logits, no learned threshold: three fixed weights, two background priors, and max operations over frozen-model outputs.

The pipeline sharpens a distribution.



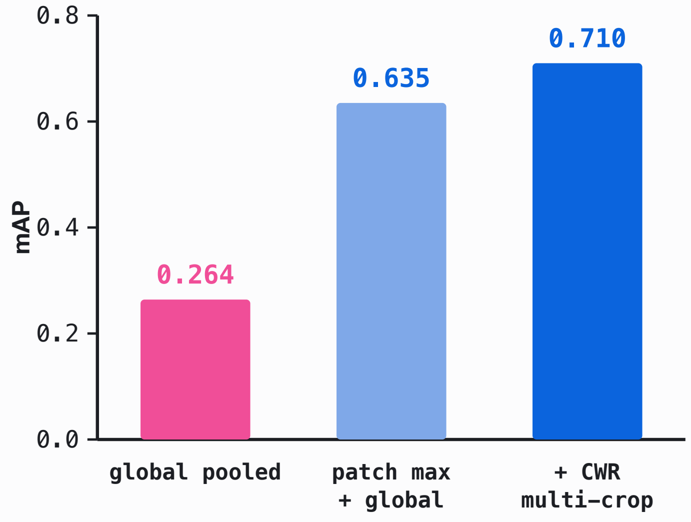
Real pipeline output, one image, all 128,260 tokens. Tagging is distribution sharpening.

RESULTS

Results on COCO-150.

METHOD	P@1	P@3	R@5	MAP
global pooled	0.433	0.289	0.449	0.264
patch max + global	0.753	0.427	0.631	0.635
softpool	0.773	0.449	0.649	0.608
+ CWR multi-crop	0.813	0.476	0.680	0.710

150 COCO-val images, real multi-label ground truth, avg 2.93 labels/image. softpool trades mAP for top-k precision; the ladder tracks the max-pool path.

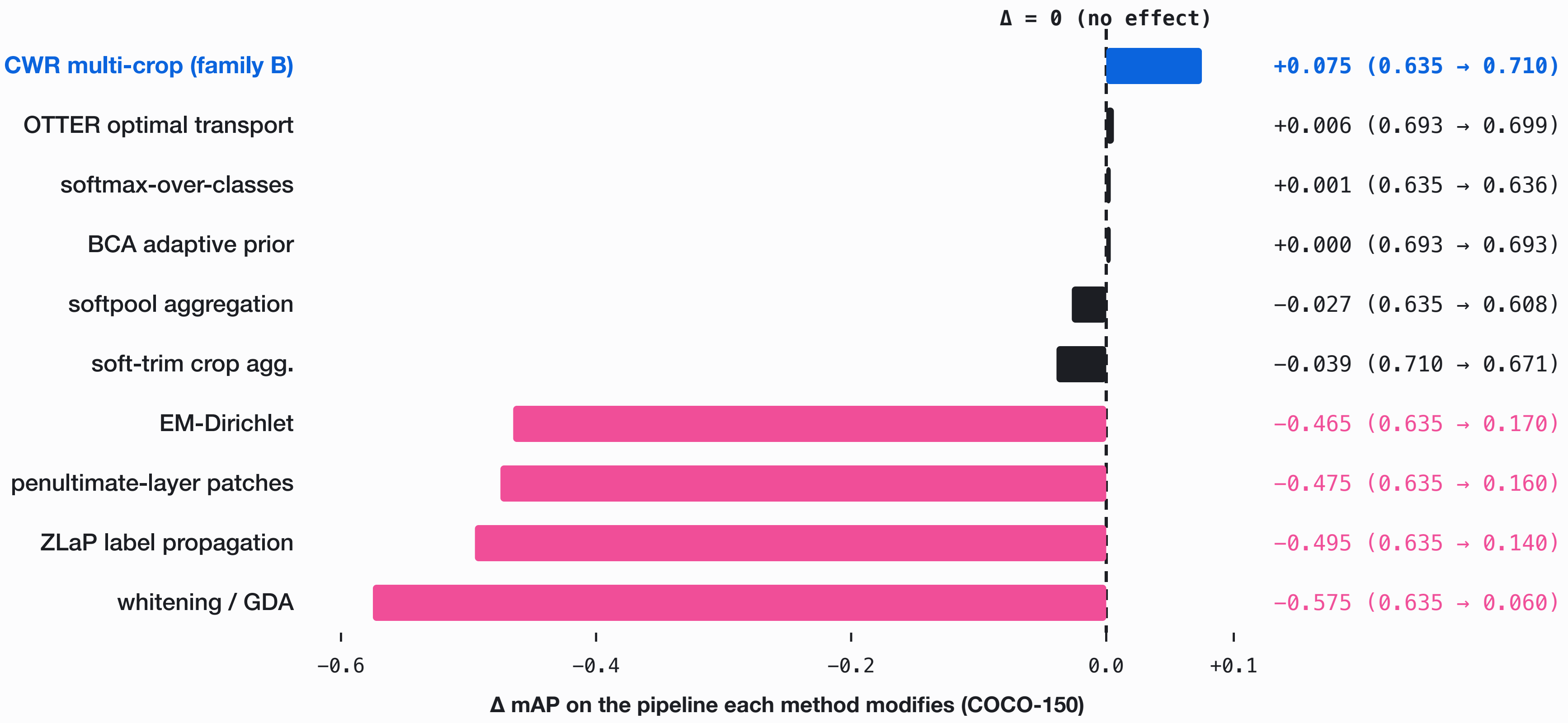


mAP: global → patch → +CWR

THE EXPERIMENT

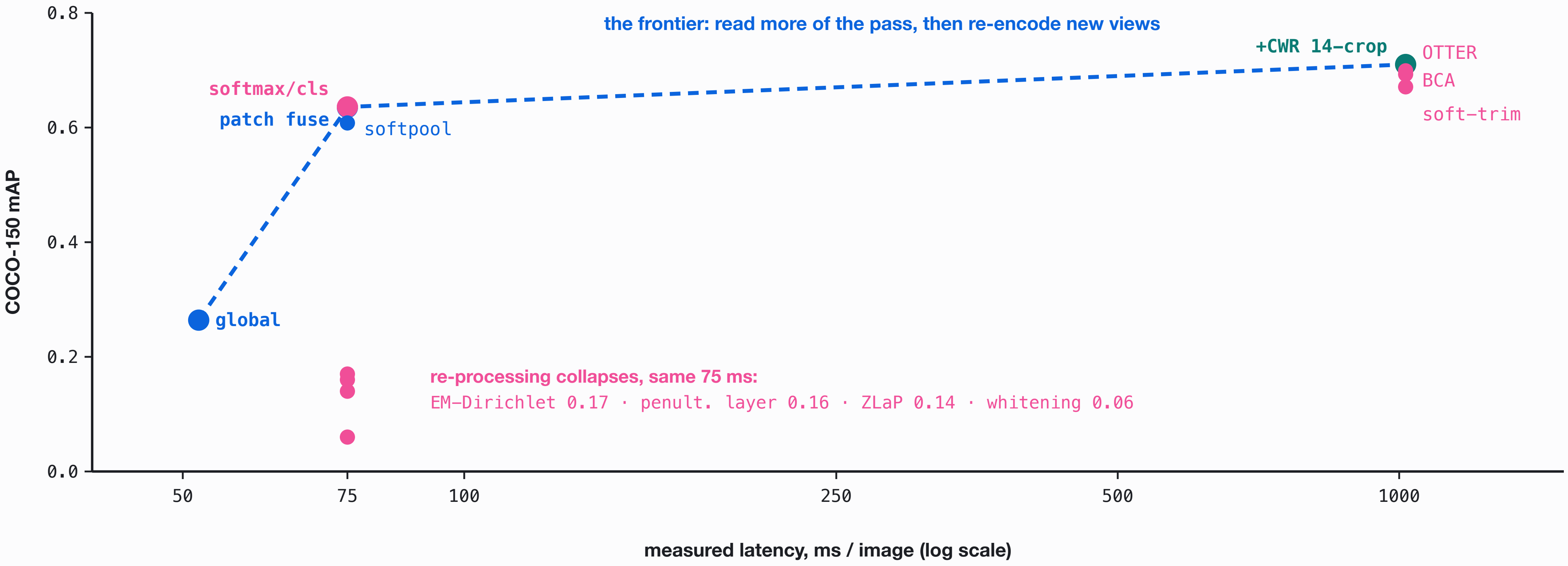
Ten training-free levers, measured.

RQ2, answered by measurement: every published lever, re-implemented on this pipeline.



THE FRONTIER

Accuracy versus test-time compute.



The frontier is traced by reading more of the pass and re-encoding new views; re-processing sits at the same cost, at or below it (post-processing adds <0.1 ms/img, measured).

ONE MORE MODE · --NGRAM

Patch-local n-grams.

the same image, the same frozen model, one extra scoring pass

--NGRAM · THE SAME FOUR TAGS, N=1..3 + BEAM

N=1 kitty · cosy · plush · crib

N=2 couch kitty · grey cosy · cat plush · sleeping crib

N=3 couch fleece kitty · grey blanket cosy · cat cloak plush · sleeping fleece crib

+BEAM grey couch kitty · sleeping black cosy · grey sleeping plush · sleeps grey crib

how the modifier is chosen · noun = kitty no word-class constraint

pool the patches where **kitty** fires → rank all 25,465 gated words on that region alone:

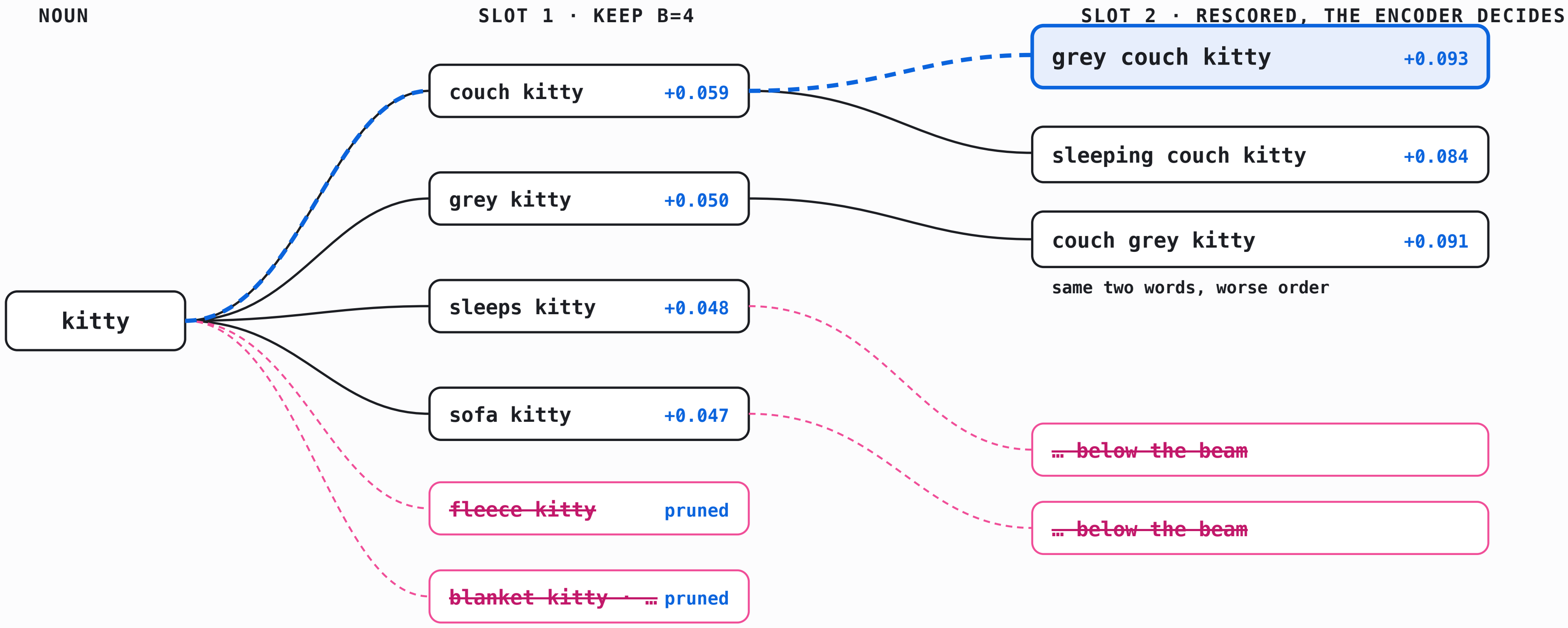
~~cat~~ · ~~kitty~~ · ~~kitten~~ (cos ≥ 0.55 to the noun: suppressed) · **couch** ✓ · fleece · blanket · grey

→ greedy picks off one ranking · --beam lets the encoder score whole phrases against the region: it promotes “grey couch kitty” and prefers attribute-first order, with no grammar anywhere

--NGRAM --BEAM · MECHANICS

Beam search over modifier slots.

Every node is a real phrase, embedded by the frozen model and scored against the noun's patch region:
$s = \langle E(\text{phrase}), \text{region}\rangle - \langle E(\text{kitty}), \text{region}\rangle$. Beam width 4.



The winning path re-orders its own words: "grey couch kitty" beats "couch grey kitty". Word order, resolved by an embedding model with no grammar.

QUALITATIVE RESULTS

Out-of-distribution photos, all three modes.

FAST: onstage · attendees · livest · concert · presenter

--HQ: onstage · venue · ceiling · projector · theater

N=2: attendees onstage · projector attendees · attendees livest · projector concert · attendees presenter

N=3: attendees projector onstage · projector onstage attendees · attendees projector livest

FAST: driv · rims · ford · traction · sidew

--HQ: carro · rims · windshield · steering · beach

N=2: windshield driv · carro rims · windshield ford · car traction · windshield sidew

N=3: windshield yacht driv · carro windshield rims · windshield yacht ford

FAST: attendees · passengers · protestors · migrants · crowded

--HQ: attendees · commuters · demonstrators · synagogue · crowded

N=2: passengers attendees · crowd passengers · crowd protestors · crowd migrants · demonstrators crowded

N=3: passengers protesters attendees · crowd protestors passengers · crowd passengers protestors

Re-encoding turns fragments into words (driv → carro, windshield); patch-local scoring attaches modifiers (n=2 top-5, n=3 top-3; all unfiltered).

THE META-CONCLUSION

Only new information moves accuracy.

Re-processing methods were designed for weakly calibrated, single-label, two-tower CLIP. This encoder's space is already aligned, calibrated, and multi-label: **there is nothing left to recover**.

RE-PROCESSING FEATURES

whitening 0.06 · ZLaP 0.14 · EM-Dirichlet 0.17 · OTTER +0.006 over its own base. None improves the pipeline it modifies.

FEEDING NEW INFORMATION

CWR crops → mAP 0.710, P@1 0.813. The only intervention that improves results.

The ceiling is the 1B model itself. Real test-time compute here means giving the model more to look at, not re-arranging what it already saw.

Versus the recent tagging literature.

LINE OF WORK	THEIR SETUP	OUR DELTA
Tag2Text / RAM / RAM++	trained tagging models, curated ~6.4k tag list	training-free, label space = tokenizer vocab
TagCLIP	CLIP two-tower, penultimate layer, DMAR+CWR	single omni model, last layer is the output space
PIAA	patch-level scoring (“[CLS] is not enough”) + GDA whitening	patch thesis confirmed (+0.37 mAP); whitening collapses (0.06)
recent calibration (OTTER, BCA)	calibration set or target prior to de-bias	one offline background prior, zero calibration

RQ1: a frozen embedding model does become a tagger, training-free. RQ2: only compute that adds information scales it. Two questions, answered.

DEPLOYMENT

OMNI-MACOS

On-device deployment in Omni.

The Python study became a feature of **Omni**, a native on-device semantic-search app (Swift + MLX, same frozen model). The port reuses the architecture directly:

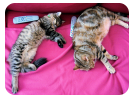
SAME FORWARD PASS	the patch rows already exist for the file's *embedding*; tagging adds one matmul.
~0.6 MS / IMAGE	about 4% overhead on the embed step. Tagging is effectively free at index time.
TAGS = SNIPPETS	tags land in the search snippet: media becomes *keyword*-searchable.
SELF-CALIBRATING	the background prior μ is estimated on-device from the first 64 images it sees, then frozen.

OMNI · ON-DEVICE SEARCH

search: carro steering

IMG_2214.jpg

carro steering rims windshield



couch-nap.heic

kitty cosy plush paw

trail-cam-04.mp4 · segment 0:00–4:00

fur bear muzzle

tag chips as they appear in results; the third row is a video segment

ONE RULE, EVERY MEDIA SHAPE

Images, video, and scanned documents.

IMAGE

Tagged at **index time**, inside the embedding forward. Patch-max + global, prior-centered, gated, NMS-deduped.

`+1 matmul · ~4% overhead`

HQ IMAGE

A background **re-tag pass** adds the production CWR variant: 5 crops, per-label max.

`P@1 0.773 → 0.847 (Swift, bf16) · 14-crop config: not deployed`

VIDEO

32 frames per **240-second segment**, one sequence through the tower: patch-max pools over **space and time**.

`one tag set per segment`

SCANNED PDF

Pages render to images, same path: **per-page tags** ("invoice, table, signature") as the snippet.

`every page, keyword-findable`

The media shape only changes what counts as a patch: crops for detail, frames for time, pages for documents.

SYNTHESIS

A capability the model was never trained for, realized at inference.

Two talks, one thesis. A frozen encoder holds more capability than its training objective exposes, and you unlock it with **test-time compute**, not more parameters.

RETRIEVAL TALK

recombine the geometry for more **relevance** on the task it was trained for.

THIS TALK

recombine the geometry for an **emergent new task** it was never trained for: tagging.

Test-time compute scales only when it supplies the model with new information. Re-processing an already-good representation yields nothing.

A frozen model already knows more than its objective admits.

Test-time compute is how you ask.

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